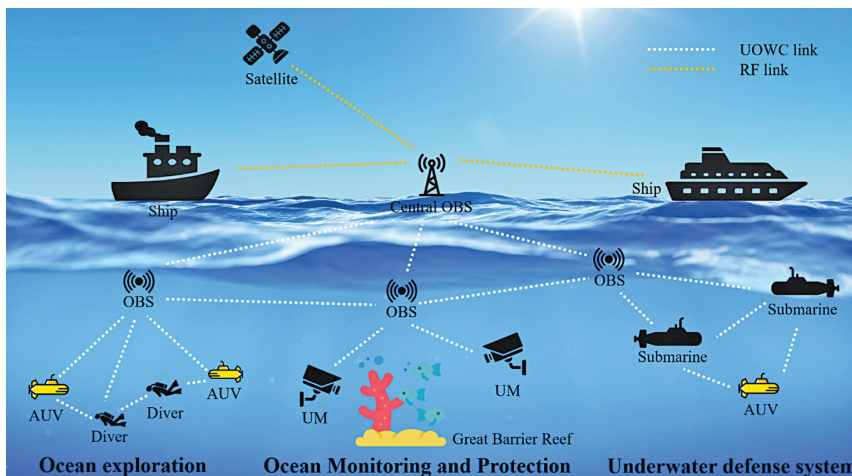
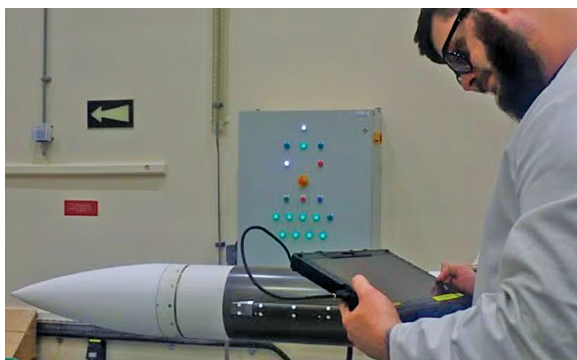




ISRO: Integrating Li-Fi into satellites for military operations (Source <https://www.indiandefensenews.in>)



Underwater optical wireless communication network architecture (Source: <https://www.mdpi.com>)



A defence equipment and support (DE&S) specialist using Li-Fi (Source: DE&S)

controlled laboratory environments as well as during field trials.

The project has demonstrated proof of concept and generated baseline performance data on latency, resilience, and spectral efficiency. This data serves as a reference for future Li-Fi research platforms and testbeds.

**Military Centre of Excellence (CoE) on Li-Fi.** The CoE, established at the Indian Institute of Information Technology, Delhi (IIIT Delhi), functions as an interface between defence agencies, academia, and industry to advance projects from laboratory concepts

to deployable military systems.

It is anchored jointly by the Telecommunications Standards Development Society India and the European Telecommunications Standards Institute to contribute to the drafting and advancement of international standardisation efforts.

The centre also develops Visible Light Communication products and technology solutions for last-mile connectivity. Examples include secure transceivers for command rooms, optical communication links for armoured vehicles providing connectivity under battlefield conditions, intra-drone communication modules for swarm coordination, and positioning systems for underground facilities where GPS signals are unavailable.

## Key global Li-Fi defence programmes

There is growing recognition that



Kitefin programme (Source: <https://www.gophotonics.com>)

light-based Li-Fi communication can strengthen operational secrecy during both war and peace. Nations across the world are exploring Li-Fi as part of their defence communication strategies.

Building on early experiments and testbeds, several countries are now focusing on deploying the technology for secure battlefield networks, satellite links, vehicular systems, and dynamic platforms such as drones, submarines, and unmanned underwater vehicles (UUVs). Some notable initiatives include:

### US Army's Kitefin programme.

A benchmark in this global transition is the US Army's Kitefin Programme, developed in collaboration with pureLiFi. Kitefin is recognised as the world's first large-scale deployment of Li-Fi, consisting of thousands of field-deployable units integrated into tactical and strategic environments.

By utilising both visible and infrared light, Kitefin provides a wireless solution with a near-zero electromagnetic signature. This makes it extremely difficult for an adversary to detect or jam a command post from outside its defined cone of light. Li-Fi links also reduce network setup time from hours to minutes, thereby improving mission readiness. The system has been validated during high-pressure manoeuvres.

**UK, Ministry of Defence - Safety in hazardous zones.** In the UK, the Ministry of Defence is deploying Li-Fi at the Defence Munitions (DM) site in Gosport, which handles complex weapon systems and explosives where conventional Wi-Fi